

Data Interpretation Exercise

Data Interpretation Assessment (R Programming)

Topics:

- Mapping Data onto Aesthetics
- Coordinate Systems and Axes
- Color Scales

Instructions: Use the given dataset to write an appropriate R program, generate the required visualization, and answer the questions.

Question 1: Mapping Data onto Aesthetics

Dataset

Product	Sales (₹ Lakhs)	Profit (%)	Category
P1	120	18	Electronics
P2	90	12	Clothing
P3	150	25	Electronics
P4	80	8	Furniture
P5	135	22	Furniture
P6	100	16	Clothing

Questions

1. Write an R program to create a scatter plot with **Sales** on the x-axis and **Profit** on the y-axis.
2. Map **Category** using an appropriate aesthetic and **Sales** using another suitable aesthetic.
3. Interpret the relationship between Sales and Profit.
4. Identify the type of scale used for each variable.
5. Suggest one alternative visualization that better represents the same data.

Question 2: Coordinate Systems (Cartesian and Polar)

Dataset

Direction	Frequency
N	18
NE	12
E	20
SE	15
S	22
SW	11
W	16
NW	14

Questions

1. Write an R program to visualize the dataset using a Cartesian coordinate system.
2. Write another R program to visualize the same data using a Polar coordinate system.
3. Compare the effectiveness of both visualizations.
4. Explain which coordinate system is more appropriate for directional data.
5. Justify your answer based on data interpretation.

Question 3: Color Scales for Data Representation

Dataset

Student	Marks
A	45
B	58
C	72

Student	Marks
D	81
E	67
F	90

Questions

1. Write an R program to create a bar chart using a sequential color scale based on Marks.
2. Identify the students with the highest and lowest marks from the visualization.
3. Explain why a sequential color scale is appropriate.
4. Suggest a color palette suitable for color-blind users.
5. Explain the difference between sequential and categorical color scales.

Question 4: Highlighting with Color

Dataset

Region	Revenue (₹ Crores)
North	820
South	760
East	790
West	1010
Central	770

Questions

1. Write an R program to create a bar chart highlighting the region with the highest revenue.
2. Explain how highlighting improves data interpretation.
3. Identify the highest-performing region.
4. Discuss how excessive use of colors can affect interpretation.
5. Suggest one alternative highlighting technique other than color.

Question 5: Integrated Visualization

Dataset

City	Population (Million)	AQI	Rainfall (mm)	Region
A	8.2	110	850	North
B	5.6	180	430	South
C	9.4	90	1200	North
D	6.8	240	520	East
E	4.9	150	680	West

Questions

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.
2. Select suitable aesthetics for Population, AQI, Rainfall, and Region.
3. Interpret the relationship between Population and AQI.
4. Identify the continuous and categorical variables.
5. Recommend one improvement to enhance the visualization's readability.